A BIDIRECTIONAL SIGN LANGUAGE INTERACTION SYSTEM USING MULTIMODAL DEEP LAERNING

CAPSTONE PROJECT REPORT

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DECLARATION

We affirm that the project work titled **"A BIDIRECTIONAL SIGN LANGUAGE INTERACTION SYSTEM USING MULTIMODAL DEEP LAERNING"** being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

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PROJECT SUMMARY

Project Title	**A BIDIRECTIONAL SIGN LANGUAGE INTERACTION SYSTEM USING MULTIMODAL DEEP LAERNING**	
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Program Concentration Area	Bidirectional Sign Language Recognition using Artificial Intelligence	
Technical Requirements	Python, OpenCV, TensorFlow/Keras, Speech Recognition, Webcam, Microphone, and Database support.	
Engineering standards and realistic constraints in these areas		
Area	**Codes & Standards / Realistic Constraints**	**Tick ✓**
Economic	The system uses open-source software and standard hardware to keep development cost low.	✓
Environmental	The project is software-based and does not cause environmental pollution or waste.	✓
Social	The system improves communication between deaf and non-signers, supporting social inclusion.	✓
Ethical	User data such as voice and video is processed securely without misuse.	✓
Health and Safety	The system is safe to use as it only requires a webcam and microphone without physical risk.	✓
Manufacturability	The software can be easily deployed on standard computers without special hardware.	✓
Sustainability	The system can be updated and reused for future improvements and additional features.	✓

ABSTRACT

Communication between Deaf and Hard-of-Hearing (DHH) individuals and people who do not understand sign language remains a significant challenge in everyday life, particularly in education, healthcare, workplaces, and public services. Traditional communication methods such as interpreters, written notes, or one-directional recognition systems are often expensive, slow, or limited in functionality. To address these limitations, this project presents BiSignAI, a bidirectional sign language interaction system designed to enable real-time communication between signers and non-signers using multimodal artificial intelligence techniques.

The proposed system provides two primary functionalities: sign-to-text/speech conversion and text/speech-to-sign generation. In the first module, hand gestures are captured through a webcam and processed using computer vision techniques. The system detects and extracts hand features using image preprocessing and landmark detection, followed by classification through deep learning models trained on Indian Sign Language (ISL) alphabet gestures. The recognized signs are converted into textual output and further synthesized into speech using a Text-to-Speech engine, enabling non-signers to understand the message both visually and audibly.

In the reverse communication pathway, users can provide input through text or speech. Speech input is first converted into text using an automatic speech recognition module. The textual data is then semantically processed using sentence embedding models to enhance contextual understanding. The system retrieves corresponding sign representations and generates animated fingerspelling sequences using pre-stored alphabet frames, allowing DHH users to receive messages in sign language form. A real-time client-server communication framework ensures smooth interaction with minimal latency.

The system is implemented using technologies such as OpenCV for gesture capture, TensorFlow/Keras for deep learning-based classification, Flask-SocketIO for real-time communication, and PostgreSQL with vector embeddings for semantic retrieval. Experimental evaluation demonstrates high responsiveness and reliable performance in real-time usage, with strong recognition accuracy for alphabet gestures and smooth animation playback. User testing indicates good usability and clarity in both recognition and generation modules.

Overall, BiSignAI provides an efficient, scalable, and accessible communication solution that bridges the interaction gap between signers and non-signers. By integrating computer vision, natural language processing, and speech technologies into a unified framework, the system offers a practical step toward inclusive communication. Future enhancements may include continuous gesture recognition, facial expression analysis, and more natural avatar-based sign synthesis to further improve expressiveness and real-world applicability.

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LIST OF ACADEMIC REFERENCE COURSES

S. NO.	COURSE CODE	COURSE NAME
1	211CSE1402	Python Programming
2	213CSE1301	Introduction to Artificial Intelligence and Machine Learning
3	213CSE2301	Predictive Analysis
4	212CSE2304	Machine Learning
5	213CSE3301	Deep Learning

CHAPTER – I

INTRODUCTION

In recent years, technological advancements have played a vital role in improving accessibility and inclusivity for individuals with disabilities. Among these, communication assistance technologies have gained significant importance, particularly for Deaf and Hard-of-Hearing (DHH) individuals who rely on sign language as their primary mode of interaction. While sign language is a rich and expressive visual language, the communication gap between signers and non-signers remains a major challenge in everyday life. This gap often limits access to education, employment, healthcare, and social participation. As a result, there is a growing need for intelligent systems that can bridge this communication barrier effectively and efficiently.

Artificial intelligence and machine learning have opened new possibilities for developing assistive technologies that can interpret human gestures, speech, and language in real time. Computer vision enables machines to recognize hand movements and gestures, while natural language processing and speech technologies allow systems to understand and generate human language. By integrating these technologies, it is now possible to create interactive systems capable of translating between sign language and spoken or written language. Such systems not only enhance accessibility but also promote independence and equal participation for DHH individuals in society.

1.1 Background and Motivation

Communication is a fundamental human need that enables interaction, learning, collaboration, and social integration. However, for Deaf and Hard-of-Hearing (DHH) individuals, communication with people who do not understand sign language remains a significant challenge. Sign languages such as Indian Sign Language (ISL), American Sign Language (ASL), and British Sign Language (BSL) are complete and expressive visual languages with their own grammar and structure. Despite their importance, a large portion of the hearing population lacks knowledge of sign language, which creates barriers in daily interactions across educational institutions, workplaces, healthcare services, and public environments.

Traditionally, communication between signers and non-signers relies on interpreters, written communication, or text-based digital tools. While interpreters provide accurate translation, they are not always available, and their services can be costly and impractical for everyday use. Written communication can be slow and ineffective, especially when conveying emotions or complex information. Existing technological solutions have attempted to bridge this gap, but most are limited to one-directional communication, primarily translating sign language into text or speech. Such systems do not support reciprocal interaction, making natural conversation difficult.

Recent advancements in artificial intelligence, particularly in computer vision, deep learning, natural language processing, and speech recognition, have opened new possibilities for developing automated and real-time communication tools. Vision-based gesture recognition systems can now detect and interpret hand movements with increasing accuracy, while speech processing and text understanding technologies allow seamless interaction through multiple

input modes. These technological developments provide an opportunity to design integrated systems that enable two-way communication between DHH individuals and non-signers.

Motivated by the need for accessible and inclusive communication, this project introduces BiSignAI, a bidirectional sign language interaction system that enables real-time translation between sign language and spoken or written language. By combining gesture recognition, speech processing, semantic understanding, and sign generation, the system aims to create a natural and efficient communication bridge that reduces dependence on interpreters and improves accessibility for DHH individuals in everyday life.

1.2 Problem Statement

Despite technological advancements, effective communication between Deaf and hearing individuals remains limited due to the lack of practical, real-time, and bidirectional translation systems. Most existing sign language recognition tools focus only on converting hand gestures into text or speech. They do not provide the reverse translation from spoken or written language into sign language, which is essential for natural two-way interaction.

Additionally, many available systems require specialized hardware such as sensor gloves or motion trackers, making them expensive and less portable. Vision-based systems often struggle with environmental variations such as lighting conditions, background clutter, and gesture similarities. Furthermore, existing solutions typically lack contextual understanding and semantic interpretation, resulting in limited conversational capability.

Therefore, there is a need for a cost-effective, real-time, and scalable system that supports both sign-to-text/speech and text/speech-to-sign conversion using widely available devices such as webcams and microphones. Such a system should be capable of understanding user input, processing it efficiently, and presenting meaningful output in an accessible form. The absence of such a comprehensive and practical solution motivates the development of the proposed bidirectional sign language communication system.

1.3 Objectives of the Project

The primary objective of this project is to develop a real-time bidirectional sign language interaction system that enables effective communication between Deaf and Hard-of-Hearing (DHH) individuals and people who do not understand sign language. The system aims to recognize sign language gestures using computer vision and deep learning techniques and convert them into readable text and audible speech to help non-signers understand the message. In addition, the project focuses on translating spoken or typed language into sign language representation through animated fingerspelling sequences, allowing DHH users to receive information in a visual and accessible format. Another important objective is to ensure real-time performance with minimal latency by integrating gesture recognition, speech processing, and semantic interpretation within a unified platform. The project also seeks to provide a user-friendly interface that supports multiple input modes, including camera, text, and voice, making the system practical for everyday use. Ultimately, the goal is to create an efficient, scalable,

and accessible communication tool that reduces communication barriers and promotes inclusivity for individuals who rely on sign language.

1.4 Scope of the Project

The scope of the proposed system focuses on enabling two-way communication using Indian Sign Language alphabet recognition and text or speech input processing. The system captures hand gestures through a standard webcam and processes them using deep learning models trained on static alphabet gestures. The recognized signs are converted into text and speech output for hearing users.

In the reverse direction, the system accepts text or voice input from non-signers and converts it into sign language representation using animated alphabet sequences. Semantic processing techniques are employed to improve contextual interpretation and response relevance.

The system is implemented as a real-time web-based application with a user-friendly interface that supports camera input, microphone input, and text input. While the current implementation focuses on alphabet-level gesture recognition and fingerspelling animation, the design allows future expansion to continuous sign recognition, facial expression analysis, and full-sentence sign generation.

1.5 Methodology Overview

The proposed system follows a multimodal processing approach that integrates computer vision, deep learning, speech processing, and natural language understanding. The methodology consists of two major functional modules: sign-to-text/speech conversion and text/speech-to-sign conversion.

In the sign recognition module, hand gestures are captured through a webcam and processed using image preprocessing techniques such as resizing, normalization, and hand region extraction. The processed frames are passed to trained deep learning models that classify the gestures into corresponding alphabet labels. The recognized output is displayed as text and converted into speech using a text-to-speech engine.

In the reverse communication module, user input is received through text or speech. Speech input is transcribed into text using an automatic speech recognition system. The text is then processed using semantic embedding techniques to understand context and retrieve corresponding sign representations. The system generates animated fingerspelling sequences using stored sign frames to visually convey the message.

The overall system operates using a real-time client-server architecture, ensuring low-latency communication and smooth interaction between users.

1.6 Organization of the Report

This report is organized into multiple chapters that describe the development and evaluation of the proposed system. Chapter I introduces the background, problem statement, objectives, and scope of the project. Chapter II presents a review of related work and existing research in sign language recognition and translation. Chapter III discusses system analysis, including requirements, feasibility, and risk assessment. Chapter IV explains the system design, architecture, modules, and user interface. Chapter V describes the implementation process and technology stack used in development. Chapter VI presents testing methodologies and evaluation results. Chapter VII discusses system performance, challenges, and comparisons with existing approaches. Finally, Chapter VIII concludes the report and outlines future enhancements for improving the system.

CHAPTER-II

LITERATURE REVIEW

Sign language recognition has developed significantly with the advancement of artificial intelligence and computer vision technologies. Early systems relied on hardware-based solutions such as sensor gloves and motion tracking devices to capture hand movements accurately, but these systems were expensive and impractical for everyday use. Later, vision-based approaches using cameras became more popular, as they allowed gesture detection without specialized equipment. Traditional image processing techniques were initially used to identify hand shapes and movements, but their performance was often affected by lighting conditions, background complexity, and variations in gesture appearance. The introduction of deep learning, particularly Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), greatly improved recognition accuracy by enabling automatic feature extraction and better handling of dynamic gesture sequences.

Recent research has focused on integrating multiple technologies such as natural language processing, speech recognition, and gesture generation to improve communication systems. Some systems translate sign language into text or speech, while others generate sign animations from spoken or written language. However, most existing solutions support only one-directional communication and lack real-time interaction and contextual understanding. These limitations highlight the need for an integrated bidirectional system that combines gesture recognition, language processing, and sign generation to enable seamless communication between sign language users and non-signers.

2.1 Overview of Related Work

Sign language recognition and translation have been widely studied as an important area of assistive technology aimed at improving communication for Deaf and Hard-of-Hearing (DHH) individuals. Early research in this domain primarily focused on sensor-based approaches, where specialized hardware such as data gloves, motion trackers, and wearable sensors were used to capture hand movements and finger positions. These systems provided accurate gesture tracking but required expensive equipment and limited user mobility, making them impractical for everyday use.

With the advancement of computer vision techniques, researchers shifted toward vision-based sign language recognition systems that rely on cameras to capture hand gestures. Traditional vision-based methods used handcrafted feature extraction techniques such as Scale-Invariant Feature Transform (SIFT), Histogram of Oriented Gradients (HOG), and contour-based detection to identify gesture patterns. Although these approaches reduced hardware dependency and improved accessibility, their performance was sensitive to variations in lighting conditions, background noise, and signer differences.

The emergence of deep learning significantly improved the performance of sign language recognition systems. Convolutional Neural Networks (CNNs) became widely used for extracting spatial features from hand images and recognizing static gestures with high accuracy. Recurrent Neural Networks (RNNs), including Long Short-Term Memory (LSTM)

models and Bidirectional LSTM architectures, were introduced to capture temporal patterns in continuous signing. These models enabled better recognition of dynamic gestures and sequences, making sign recognition more robust and scalable.

Recent research has also explored multimodal approaches that combine computer vision, natural language processing, and speech technologies. Transformer-based models, such as BERT and Sentence-BERT, have been used to enhance contextual understanding and semantic interpretation of textual data. Additionally, gesture recognition systems have been integrated with real-time communication frameworks to support interactive applications. Despite these advancements, many systems remain limited to one-directional translation and lack complete bidirectional communication capabilities.

2.2 Review of Similar Projects or Research Papers

Several research studies and practical systems have contributed to the development of automated sign language recognition and translation technologies. Early large-scale recognition systems focused on continuous sign language interpretation using hybrid models that combined convolutional networks with probabilistic sequence models. These approaches demonstrated the feasibility of recognizing complex gesture sequences but often required extensive training data and computational resources.

End-to-end neural translation models have also been proposed to convert sign language videos directly into spoken language sentences. These models utilize deep learning architectures to learn relationships between visual gestures and linguistic meaning. Although effective in structured environments, such systems often struggle with real-time processing and natural interaction.

In the area of sign language generation, some researchers have developed avatar-based or animation-based systems that produce visual sign representations from textual input. Rule-based systems using symbolic representations were initially used to generate sign movements, but they often produced rigid and unnatural gestures. More recent work has explored motion generation using deep learning and generative models to create smoother and more realistic animations. However, accurately representing facial expressions, body posture, and emotional cues remains a significant challenge.

Multimodal systems that integrate visual, textual, and auditory data have shown promising results in improving recognition accuracy and contextual understanding. Some systems incorporate semantic embeddings and vector-based retrieval methods to enhance response relevance. Real-time communication frameworks and lightweight models have also been developed to enable deployment on mobile or web platforms.

Despite these developments, many existing solutions focus either on recognition or generation but not both. Fully integrated systems that support bidirectional translation, real-time performance, and practical usability remain relatively limited, particularly for Indian Sign Language (ISL), which lacks large annotated datasets and standardized digital resources.

2.3 Summary and Gap Identification

The literature review reveals that significant progress has been made in sign language recognition, translation, and generation through advancements in computer vision and deep learning. Sensor-based systems provided early accuracy but lacked practicality due to hardware dependency. Vision-based systems improved accessibility but faced environmental limitations. Deep learning models enhanced recognition accuracy and enabled dynamic gesture interpretation, while multimodal approaches improved contextual understanding and response relevance.

However, several research gaps remain. Many existing systems provide only one-directional communication, typically converting sign language into text or speech without supporting the reverse process. Systems capable of generating sign language from spoken or written input often rely on simplified or unnatural animation techniques. Real-time performance and low-latency interaction are also major challenges, particularly in resource-constrained environments. Additionally, limited availability of large-scale datasets for Indian Sign Language restricts the development of highly accurate and comprehensive models.

To address these gaps, the proposed system integrates gesture recognition, speech processing, semantic analysis, and sign generation within a unified real-time framework. By supporting both sign-to-text/speech and text/speech-to-sign translation, the system provides complete bidirectional communication. The use of deep learning for gesture classification, semantic embeddings for contextual understanding, and animation-based sign synthesis offers a practical and scalable solution for accessible communication. This integrated approach distinguishes the proposed system from existing one-directional or hardware-dependent solutions and contributes to the development of inclusive assistive technology.

CHAPTER III

SYSTEM ANALYSIS

System analysis is an essential phase in software development that involves identifying system requirements, evaluating feasibility, and understanding potential risks before implementation. The purpose of this chapter is to analyze the functional and non-functional requirements of the proposed bidirectional sign language interaction system, evaluate its feasibility, and identify challenges that may arise during development and deployment. The analysis ensures that the system is practical, efficient, and capable of meeting user needs.

In addition to identifying requirements and evaluating feasibility, system analysis also focuses on ensuring that the proposed solution meets real-world communication needs effectively. The bidirectional sign language interaction system must handle different user environments, variations in gesture performance, and diverse communication scenarios. Therefore, careful attention is given to system performance, reliability, and adaptability during the analysis phase. The system is designed to function under varying lighting conditions, different user hand positions, and diverse speech inputs while maintaining consistent accuracy. This phase also emphasizes scalability, allowing the system to support future enhancements such as continuous gesture recognition, multilingual support, and more natural sign generation methods. By thoroughly analyzing these factors, the system is structured to provide efficient, reliable, and practical communication support for both sign language users and non-signers.

3.1 Requirements Gathering

Requirements gathering is the process of identifying the needs and expectations of users and determining the resources required to develop the system. The proposed system is designed to assist communication between Deaf and Hard-of-Hearing (DHH) individuals and non-signers through real-time sign language translation.

The system requires a webcam to capture hand gestures, a microphone for speech input, and a computing device capable of running deep learning models. Software requirements include machine learning libraries for gesture recognition, speech processing tools for voice input, and a database system for storing semantic information and sign representations. The system also requires a user interface that allows users to interact through camera, text, and voice inputs. In addition, a stable internet or local server environment is necessary to support real-time communication between system components.

3.2 Functional Requirements

Functional requirements describe the specific operations that the system must perform. The proposed system must capture hand gestures through a camera and process them using computer vision techniques. It should classify recognized gestures into corresponding sign language alphabets or words using trained deep learning models. The system must convert recognized signs into readable text and generate speech output using a text-to-speech engine.

In the reverse direction, the system should accept text or voice input from users. Speech input must be converted into text using an automatic speech recognition module. The system must process textual input semantically and generate corresponding sign language representation through animated fingerspelling sequences. The system should provide real-time interaction with minimal delay and display outputs clearly through a user-friendly interface. Additionally, it should store and retrieve relevant data for efficient processing and response generation.

S NO	FUNCTIOAL REQUIREMETS	MODULES
1	Capture hand gestures using webcam.	Sign Recognition Module
2	Detect and classify sign language gestures	Gesture Classification Module
3	Convert recognized signs into text	Sign-to-Text Module
4	Accept speech input from user and convert to sign	Speech Recognition Module
5	Process text and map it to corresponding sign representation	Semantic Processing Module
6	Generate animated sign language output from text or speech	Sign Animation Module
7	Display real-time outputs through user interface	User Interface Module

3.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance standards of the system. The proposed system must provide real-time response with minimal latency to ensure smooth communication between users. It should maintain high recognition accuracy under different lighting conditions and backgrounds. The system must be reliable and capable of operating continuously without failure.

Usability is an important requirement, as the interface should be simple and easy to understand for both sign language users and non-signers. The system should also be scalable, allowing future integration of additional gestures, languages, or features. Security and privacy must be maintained, particularly when handling user voice data or stored information. The system should be efficient in resource usage so that it can operate on standard computing devices without requiring specialized hardware.

3.4 Feasibility Study

A feasibility study evaluates whether the proposed system can be successfully developed and implemented within available resources and constraints. It examines technical, operational, and economic feasibility.

3.4.1 Technical Feasibility

The system is technically feasible because the required technologies, such as computer vision, deep learning frameworks, speech recognition tools, and web-based communication platforms, are readily available and well-supported. Open-source libraries and pretrained models can be used to implement gesture recognition and language processing efficiently. Standard hardware such as webcams and microphones is sufficient for system operation, making the technical requirements achievable.

3.4.2 Operational Feasibility

The system is operationally feasible as it is designed with a user-friendly interface that allows easy interaction through camera, text, or voice input. Users do not need specialized training or equipment to operate the system. The application can be used in various environments, including educational institutions, workplaces, and public service settings. Its ability to support real-time communication ensures practical usability.

3.4.3 Economic Feasibility

The system is economically feasible because it does not require expensive hardware such as sensor gloves or motion capture devices. Most software components are open-source, reducing development costs. The system can run on standard computers or laptops, making it affordable for institutions and individuals. Maintenance and operational costs are also minimal, making the solution cost-effective.

3.5 Risk Analysis

Risk analysis identifies potential challenges that may affect system performance or implementation. One major risk is variation in lighting conditions or background environments, which may affect gesture recognition accuracy. Another risk is the similarity between certain hand gestures, which may lead to misclassification. Hardware limitations such as low-resolution cameras or poor audio quality may also impact system performance.

There is also a risk of latency if the system processes large amounts of data in real time without optimization. User-related factors, such as improper hand positioning or unclear speech input, may affect recognition accuracy. To minimize these risks, preprocessing techniques, model optimization, and user guidance mechanisms can be implemented. Continuous testing and system updates can further improve reliability and performance.

CHAPTER IV

SYSTEM DESIGN

System design is a crucial phase in the software development lifecycle that defines the structure, components, data flow, and operational behavior of the proposed system. It transforms the requirements identified during system analysis into a structured framework that guides implementation. A well-designed system ensures efficient processing, scalability, reliability, and user-friendly interaction.

The proposed bidirectional sign language interaction system is designed to enable real-time communication between Deaf and Hard-of-Hearing (DHH) individuals and non-signers. The system integrates computer vision, deep learning, speech recognition, and natural language processing within a unified architecture. It supports multiple input modes, processes data efficiently, and produces meaningful outputs in both textual and visual forms. The system is structured using modular design principles to improve flexibility, maintainability, and performance.

4.1 Overall System Architecture

The overall system architecture of the proposed bidirectional sign language interaction system is designed to enable real-time communication between Deaf and Hard-of-Hearing (DHH) individuals and non-signers. As shown in the block diagram, the system is organized into three main stages: input acquisition, processing, and output generation. In the input stage, the system collects data from multiple sources, including a webcam for capturing hand gestures, a microphone for receiving speech input, and a keyboard for text entry. These inputs are forwarded to the processing layer, where different computational modules analyze and interpret the information using computer vision, deep learning, and natural language processing techniques.

In the processing stage, gesture input is preprocessed to detect the hand region and extract features, which are then classified using trained deep learning models to identify the corresponding sign language gesture. The recognized gesture is converted into text and further transformed into speech for non-signers. In the reverse direction, text or speech input is converted into textual form and processed semantically to determine its meaning. The system retrieves the corresponding sign representation from stored data and generates animated sign language sequences for visual communication. The architecture also includes a database for storing semantic mappings and animation references, along with a real-time communication framework that ensures smooth interaction and low-latency data exchange between system components.

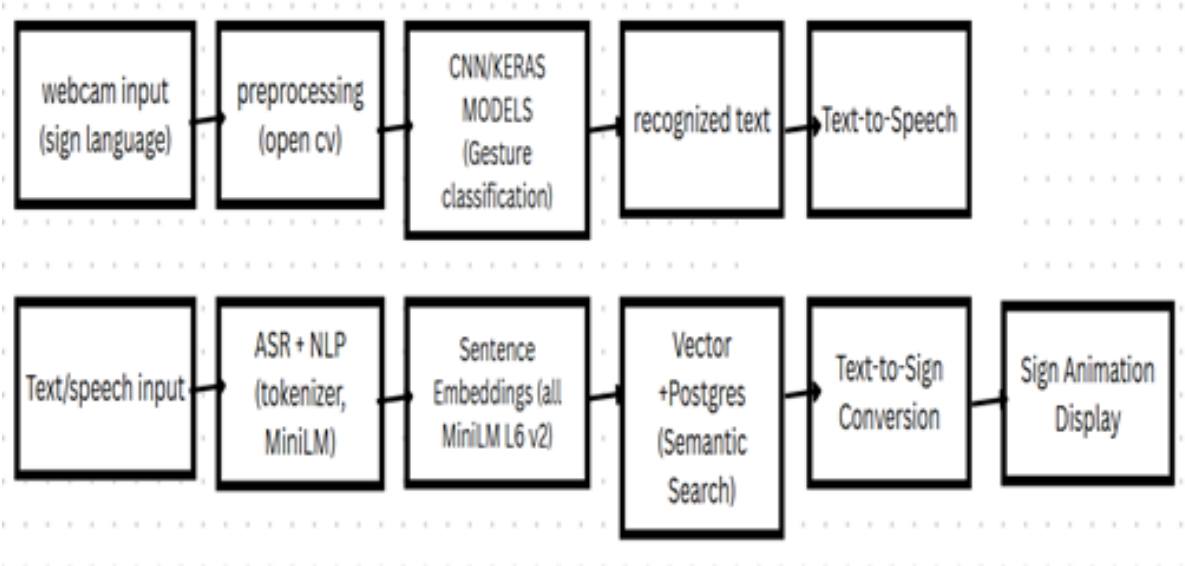

Fig 1 : System Architecture

4.2 Module Design

The system is divided into functional modules, each responsible for a specific task. Modular design improves system organization and allows independent development and testing of components.

4.2.1 Module 1 – Gesture Capture and Preprocessing Module

This module captures real-time video frames using a webcam and prepares them for gesture recognition. The captured frames are processed to improve image quality and isolate the hand region. Preprocessing operations include resizing, normalization, and background filtering.

Hand landmark detection is performed to identify key points of the hand such as finger joints and tips. These landmarks represent the spatial structure of the hand and help in accurate gesture recognition. The hand landmark detection process is illustrated in Figure 2, which shows the 21 keypoints used to represent hand structure.

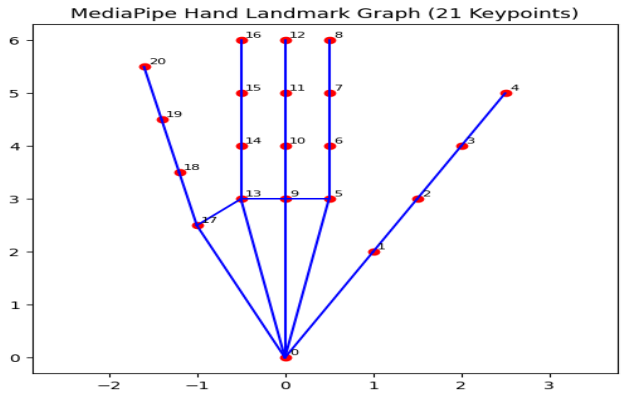

Fig 2 : MediaPipe Hand Landmark Graph

4.2.2 Module 2 – Gesture Recognition and Text Conversion Module

This module identifies sign language gestures and converts them into text. The processed hand features are passed to trained deep learning models that classify gestures into corresponding sign language alphabets or words.

Once the gesture is recognized, the system converts it into textual output and displays it on the user interface. This module enables sign language users to communicate their message in textual form for non-signers.

4.2.3 Module 3 – Speech Recognition Module

This module captures speech input from the user through a microphone. The audio signal is processed using automatic speech recognition technology to convert spoken language into text.

The recognized text serves as input for the sign generation module, enabling non-signers to communicate through sign language output.

4.3 User Interface Design

The user interface of the proposed bidirectional sign language interaction system is designed to provide a simple, intuitive, and accessible communication platform for both sign language users and non-signers. The interface supports multiple input modes and provides real-time output to ensure smooth interaction between users.

The main interface consists of three primary components: gesture input interface, speech input interface, and output display area. The gesture input interface allows users to activate the webcam and perform sign language gestures. The system captures video frames and provides visual feedback during gesture detection to guide the user. The speech input interface enables users to record voice input using a microphone. Once the speech is captured, the system processes it automatically without requiring additional user actions.

The output display area presents the system response in an easily understandable form. When gestures are recognized, the corresponding text is displayed clearly on the screen. When speech input is processed, the system generates and displays sign language animation for visual communication. The interface layout ensures that users can easily switch between gesture and speech input modes.

The design emphasizes usability and accessibility. Large display areas, clear visual indicators, and minimal control options make the system easy to operate for users with different technical backgrounds. The interface ensures real-time response, providing immediate feedback to maintain natural communication flow.

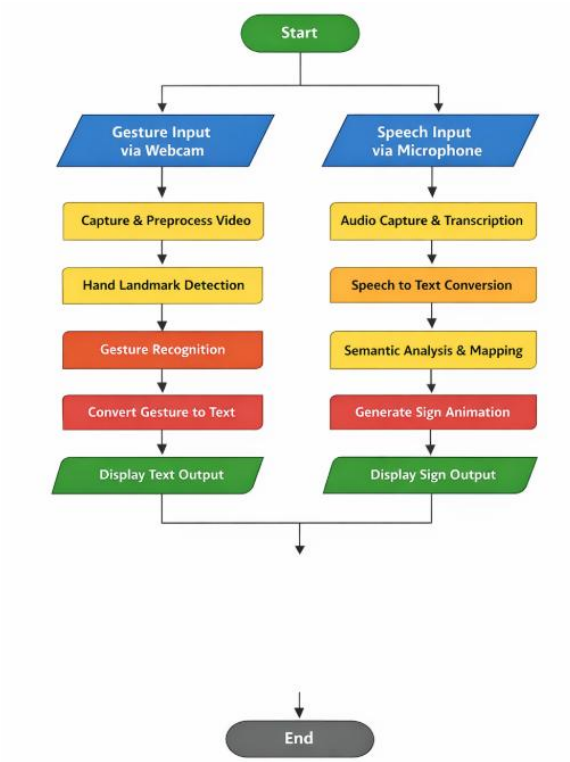

Fig 3 : Flowchart

4.3.1 User Flow Diagrams

User flow diagrams represent the sequence of actions performed by the user and the system during interaction. These diagrams help illustrate how input is captured, processed, and converted into meaningful output.

In the gesture-to-text communication flow, the user activates the webcam and performs a sign gesture. The system captures the video, processes the hand region, detects hand landmarks, and classifies the gesture using trained deep learning models. The recognized gesture is then converted into text and displayed on the interface.

In the speech-to-sign communication flow, the user provides voice input through the microphone. The system records the audio and converts it into text using speech recognition. The processed text is mapped to stored sign representations, and animated sign sequences are generated and displayed visually.

These user flows ensure smooth bidirectional communication by allowing users to interact naturally through either gestures or speech. The structured flow of operations helps maintain real-time processing and ensures consistent system performance.

CHAPTER V

IMPLEMENTATION

Implementation is the phase where the designed system is developed and integrated into a functional application. This stage involves selecting appropriate technologies, developing individual modules, and combining them into a unified system that performs real-time bidirectional communication. The proposed system is implemented using computer vision, deep learning, and speech processing technologies to enable interaction between sign language users and non-signers.

5.1 Technology Stack

The implementation of the system requires a combination of software tools, programming frameworks, and libraries that support real-time processing and machine learning operations. The technology stack is selected based on performance, scalability, and compatibility with computer vision and speech processing tasks.

The backend of the system is developed using Python due to its extensive support for machine learning and image processing libraries. Computer vision tasks such as video capture and hand detection are implemented using OpenCV. Deep learning models used for gesture recognition are developed and executed using TensorFlow and Keras frameworks. Speech input processing is performed using automatic speech recognition libraries that convert audio signals into text.

The system interface is developed using web-based technologies to provide real-time interaction. A client-server communication framework ensures continuous data exchange between frontend and backend modules. Data storage is handled using a structured database that maintains sign mappings and animation references required for sign generation.

5.1.1 Programming Languages and Tools

The proposed bidirectional sign language interaction system is developed using a combination of programming languages and software tools that support real-time processing, computer vision, and machine learning operations. The primary programming language used for system development is Python, which provides a flexible and efficient environment for implementing artificial intelligence and data processing tasks. Python is widely used in machine learning applications due to its simplicity, extensive library support, and strong community resources, making it suitable for developing gesture recognition and speech processing modules.

For image and video processing, the system uses OpenCV, an open-source computer vision library that enables real-time video capture, frame processing, and hand detection. OpenCV plays a crucial role in capturing webcam input and preparing image data for gesture recognition. The deep learning models used for sign classification are implemented using TensorFlow and Keras, which provide powerful tools for building, training, and deploying neural network models. These frameworks support efficient feature extraction and classification of hand gestures.

Speech processing is handled using automatic speech recognition libraries, which convert spoken audio into text for further processing. These libraries analyze sound signals and

generate accurate textual representations of speech input. For system interaction and real-time communication, a web-based framework is used to connect frontend and backend components, allowing users to provide input and view outputs seamlessly. Additionally, a structured database system is used to store sign mappings, gesture labels, and animation references required for sign generation. Together, these programming languages and tools provide the technical foundation necessary to implement a reliable and efficient real-time communication system.

5.2 Implementation of Modules

The implementation of the proposed system is carried out by dividing the entire functionality into separate modules, each responsible for a specific task. This modular approach improves system organization, simplifies development, and allows each component to operate efficiently while maintaining real-time communication.

The gesture capture and preprocessing module is responsible for acquiring real-time video input from the webcam and preparing it for analysis. The captured frames are processed to improve image quality and isolate the hand region from the background. Preprocessing techniques such as resizing, normalization, and noise filtering are applied to ensure consistent input for gesture recognition. Hand landmark detection is then performed to extract key spatial features that represent finger positions and hand structure.

The gesture recognition module analyzes the extracted hand features and classifies them into corresponding sign language alphabets. This module uses trained deep learning models to identify patterns in hand movements and shapes. The predicted output is converted into readable text and displayed on the user interface. The recognition process runs continuously to provide real-time response during gesture interaction.

The speech recognition module processes audio input from the user. The system captures speech through a microphone and converts the audio signal into textual form using automatic speech recognition techniques. The generated text is then forwarded to the sign generation module for visual representation.

The sign generation module converts textual input into sign language animation. The system maps the recognized text to stored gesture data and generates animated sign sequences using predefined frames. These frames are arranged sequentially to produce smooth visual motion, which is displayed on the user interface for sign language users.

The data management module handles storage and retrieval of system information. It maintains mappings between text and sign representations and stores animation references required for output generation. Efficient data retrieval ensures that the system produces accurate results with minimal delay.

Finally, the user interface module provides a platform for interaction between the user and the system. It allows users to capture gestures, provide speech input, and view results in real time. The interface is designed to be simple and accessible, ensuring smooth operation and effective communication.

Together, these modules work in coordination to enable real-time bidirectional communication, ensuring accurate gesture recognition and clear sign generation.

S.No	Module Name	Description of Implementation	Technologies Used
1	Gesture Capture and Preprocessing Module	Captures real-time video from webcam and processes frames by resizing, filtering noise, and detecting hand region for further analysis.	OpenCV, Python
2	Gesture Recognition Module	Extracts hand landmarks and classifies gestures into sign language alphabets using trained deep learning models.	TensorFlow, Keras
3	Speech Recognition Module	Captures speech input through microphone and converts audio signals into text using speech recognition algorithms.	Speech Recognition Library, Python
4	Sign Generation Module	Converts processed text into visual sign language animation using stored gesture frames and displays output.	Animation Processing, Python
5	Data Storage and Retrieval Module	Stores gesture mappings, text-sign relationships, and animation data for efficient retrieval and processing.	Database System
6	User Interface Module	Provides interface for gesture input, speech input, and output display, enabling real-time interaction.	Web Technologies, Python

5.2.1 Module 1 – Gesture Capture and Recognition

This module captures real-time video using a webcam and processes frames for gesture recognition. The captured frames are preprocessed by resizing, filtering noise, and isolating the hand region. Hand landmark detection is applied to identify key points of the hand such as finger joints and tips.

The extracted features are passed to trained deep learning models that classify the gesture into the corresponding sign language alphabet. The predicted label is converted into text and displayed on the interface. The module operates continuously to provide real-time gesture recognition.

5.2.2 Module 2 – Speech Processing

This module captures speech input from the microphone and converts it into textual form. The recorded audio signal is processed using automatic speech recognition algorithms that analyze sound patterns and generate corresponding text output.

The recognized text is then forwarded to the sign generation module for visual representation. This module enables non-signers to communicate using speech input.

5.2.3 Module 3 – Sign Generation

This module converts processed text into visual sign language output. The system maps textual input to stored sign animation frames. These frames are combined sequentially to generate smooth sign animation.

The generated animation is displayed on the user interface, allowing sign language users to understand the spoken message visually. The module ensures that animations are displayed clearly and in real time.

5.3 Integration of Modules

After implementing individual modules, they are integrated to form a complete system. The integration process ensures smooth data flow between gesture recognition, speech processing, and sign generation components.

When a user performs a gesture, the gesture recognition module processes video input and sends the recognized label to the output interface. When speech input is provided, the speech processing module converts audio to text and forwards it to the sign generation module. The generated output is then displayed visually.

A real-time communication mechanism connects frontend and backend components, allowing continuous interaction without delay. Proper synchronization between modules ensures accurate processing and consistent system performance.

CHAPTER VI

TESTING

Testing is an essential phase in system development that ensures the proposed application functions correctly, efficiently, and reliably under different operating conditions. The objective of testing is to identify errors, verify system performance, and confirm that all modules work together as intended. For the proposed bidirectional sign language interaction system, testing is performed to evaluate gesture recognition accuracy, speech processing reliability, system response time, and overall usability.

The system is tested at multiple levels, including unit testing, integration testing, system testing, and user acceptance testing. Each level focuses on verifying specific components and ensuring smooth communication between modules.

6.1 Testing Methodology

The testing methodology followed for the system is based on a structured approach that verifies individual modules and their combined functionality. Each module is tested independently before integration to ensure proper operation. After integration, the complete system is tested in real-time scenarios to evaluate performance and reliability.

The testing process involves providing gesture and speech inputs under different environmental conditions, such as variations in lighting, background, and speech clarity. The system output is observed and compared with expected results to measure accuracy and consistency.

6.1.1 Unit Testing

Unit testing is performed to verify the functionality of individual modules independently. Each module is tested with sample inputs to ensure correct processing and output generation.

The gesture capture module is tested to confirm that the webcam captures video frames correctly and detects hand regions. The gesture recognition module is tested using predefined gesture samples to verify classification accuracy. The speech recognition module is tested with different voice inputs to ensure accurate text conversion. The sign generation module is tested by providing text input and verifying that the correct sign animation is displayed.

Unit testing ensures that each component functions properly before system integration.

6.1.2 Integration Testing

Integration testing is performed after combining individual modules to ensure proper communication between them. The interaction between gesture recognition and text output modules is tested to verify seamless conversion from sign to text. Similarly, the speech recognition and sign generation modules are tested together to ensure correct mapping of speech input to sign animation.

This testing stage verifies that data flows correctly between modules and that system responses are generated without delay or error.

6.1.3 System Testing

System testing evaluates the performance of the complete system under real operating conditions. The system is tested with live gesture input and real speech input to verify end-to-end functionality. Testing is performed in different lighting environments and with different speech variations to ensure reliability.

System testing also measures response time and checks whether the system produces outputs in real time. The overall usability of the system interface and output clarity are also evaluated during this phase.

6.1.4 User Acceptance Testing (UAT)

User acceptance testing is conducted to verify that the system meets user expectations and practical communication needs. Test users interact with the system by performing gestures and providing speech input. They evaluate system accuracy, response time, and ease of use.

Feedback from users is collected to identify areas for improvement and ensure that the system provides a comfortable and effective communication experience.

6.2 Test Cases and Results

Several test cases are executed to verify system functionality. Each test case includes input conditions, expected output, and observed results. Gesture recognition tests confirm whether specific hand signs produce correct text output. Speech processing tests verify whether spoken words generate appropriate sign animations.

The results indicate that the system performs reliably in standard operating conditions. Gesture recognition accuracy remains consistent under proper lighting, and speech input is successfully converted into sign representation with minimal delay. Minor variations in lighting and background may slightly affect recognition accuracy, but overall system performance remains stable.

6.3 Bug Tracking and Resolution

During testing, minor issues related to gesture misclassification and speech clarity were identified. Some gestures with similar hand shapes occasionally produced incorrect predictions, especially under poor lighting conditions. These issues were addressed by improving preprocessing techniques and refining model parameters.

Speech recognition errors caused by unclear pronunciation or background noise were minimized by adjusting audio processing settings and filtering noise. Continuous testing and debugging helped improve system accuracy and reliability.

CHAPTER VII

RESULTS AND DISCUSSION

The results and discussion chapter presents the performance of the proposed bidirectional sign language interaction system. The system is evaluated based on its ability to accurately recognize gestures, convert speech into sign animations, and provide real-time responses. The results demonstrate the effectiveness of integrating computer vision and speech processing for assistive communication.

7.1 System Output Screenshots

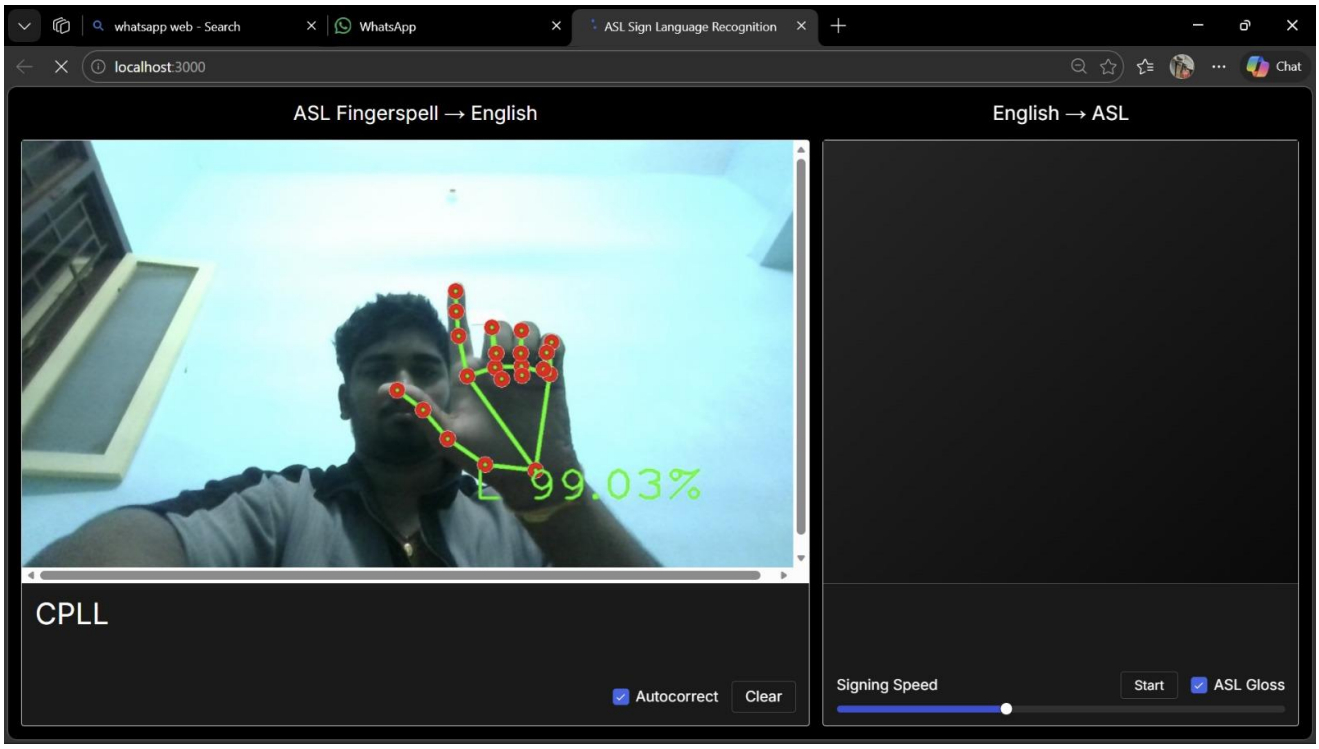

Fig 4: Displays the input sign and converts it into text.

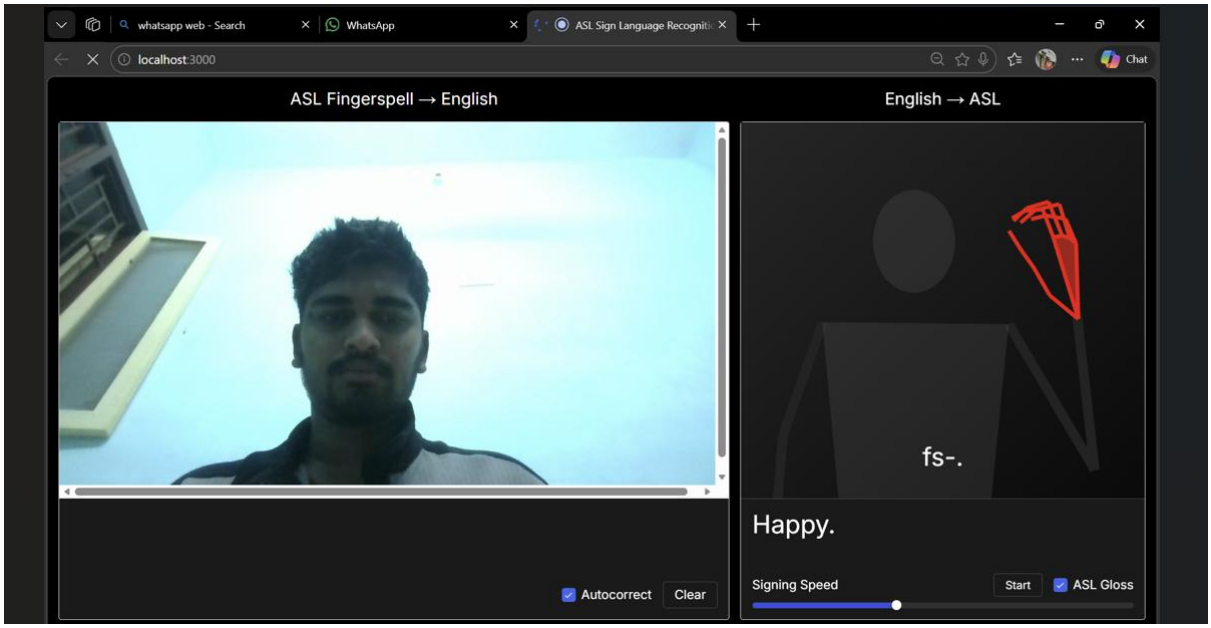

Fig 5: The given speech is converted into text and then translated into sign.

7.2 Evaluation Metrics

The performance of the system is evaluated using key metrics such as accuracy, response time, and usability. Gesture recognition accuracy measures how correctly the system identifies sign language gestures. The system achieves high accuracy for most alphabet gestures under proper lighting conditions.

Response time is another important metric, as the system is designed for real-time communication. The system provides output with minimal delay, ensuring smooth interaction between users. Usability is evaluated based on ease of use, interface clarity, and user experience. The system performs efficiently and provides a user-friendly interface for both sign language users and non-signers.

7.3 Comparison with Existing Systems

Compared to traditional sign language recognition systems, the proposed system offers improved functionality by supporting bidirectional communication. Many existing systems focus only on converting sign language into text or speech, whereas this system also supports speech-to-sign conversion.

Additionally, the system uses standard hardware such as webcams and microphones, eliminating the need for expensive devices like sensor gloves. The real-time processing capability and integrated design provide better usability and accessibility compared to existing solutions.

7.4 Challenges Faced

During system development, several challenges were encountered. One major challenge was the similarity between certain hand gestures, which sometimes led to incorrect classification. Variations in lighting conditions and background environments also affected gesture detection accuracy.

Another challenge was handling speech input with background noise or unclear pronunciation, which occasionally impacted text conversion accuracy. Ensuring real-time performance while processing both video and audio data simultaneously was also a significant challenge.

7.5 Solutions and Improvements

To overcome these challenges, preprocessing techniques were improved to enhance image quality and reduce noise in gesture recognition. Data augmentation and model tuning were applied to improve classification accuracy for similar gestures.

Speech recognition performance was enhanced by applying noise filtering techniques and optimizing audio processing. The system was also optimized for faster processing to reduce latency and improve real-time performance.

Further improvements can include expanding the dataset, improving gesture recognition models, and enhancing sign animation quality for better communication.

CHAPTER VIII

CONCLUSION & FUTURE SCOPE

The proposed bidirectional sign language interaction system provides an effective and practical solution for communication between Deaf and Hard-of-Hearing (DHH) individuals and non-signers. The system supports two primary interaction modes: converting sign language gestures into text and converting spoken language into visual sign representation. By integrating computer vision, deep learning, and speech recognition technologies, the system enables real-time communication without requiring specialized hardware or human interpreters. The gesture recognition module accurately captures hand movements through a webcam and converts them into text, while the speech processing module converts spoken input into sign language animation for visual understanding. The modular design, real-time processing capability, and user-friendly interface make the system suitable for use in educational institutions, workplaces, and everyday communication. Overall, the system successfully reduces communication barriers and contributes to improved accessibility and inclusivity for sign language users.

Although the system performs effectively, several enhancements can further improve its functionality and performance. Currently, the system mainly supports alphabet-level or basic gesture recognition; future work can extend it to continuous sign recognition for full words and sentences. The sign generation module can also be enhanced with more realistic animations or 3D avatars that include facial expressions and body movements, making communication more natural and expressive. Improvements in speech recognition accuracy, especially in noisy environments, and support for multiple spoken and sign languages can expand usability. Future development may also include mobile application deployment, integration with video communication platforms, and support for additional assistive features. With these improvements, the system can evolve into a more comprehensive and widely accessible communication tool for inclusive interaction across various social and professional environments.

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PUBLICATION

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Acceptance Letter

Paper ID	Paper Title	Author's Name
ICECA-298	A Bidirectional Sign Language Interaction System using Multimodal Deep Learning	M K Nagarajan, Sadda Krishna Teja, NoorBasha Moushid, Chavala Sarath Kumar, Nakka Venkata Bhanu Prakash

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Abstract

Document Sections

I. INTRODUCTION

II. LITERATURE REVIEW

III. METHODOLOGY

IV. RESULTS AND DISCUSSIONS

Abstract:
Deaf and Hard-of-Hearing (DHH) individuals often encounter communication challenges when interacting with people who don't know sign language. Traditional solutions like interpreters or one-way recognition tools do not offer scalable and real-time access. This work introduces BiSignAI, a two-way sign language interaction system built with computer vision, natural language processing, and speech technologies. The system uses FlaskSocketIO for real-time communication, OpenCV for capturing gestures, and several pre- trained deep learning models (Keras/TensorFlow) for sign recognition. For processing text and speech, sentence-transformers (all- MiniLML6-v2) and LLM-based modules manage embeddings and contextual interpretation, while pgvector combined with PostgreSQL allows for effective semantic search and storage. In the other direction, BiSignAI translates text into animated A-Z fingerspelling sequences using pre-stored alphabet frames. The prototype shows high responsiveness, a scalable setup, and practical usability, enhancing

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